PIC Finance Module Analysis: M10 Invoice Matching & Finance Integration (flow F7), finance parts of M11

Scope: Blueprint M10 (section 16) and flow F7 (section 21.7), the finance parts of M11 (finance validation, budget impact, SOD-07), and Technical Document payables-svc (sections 5, 6, 7.3 DMN-03, 9.4, 10, 11, 12.3, 14). Code: `backend/app/pic/services/payables.py` and its neighbours, `frontend/src/app/pic/invoices`. Method: read the specifications and every code path, ran the existing finance tests (baseline 108 passed), fixed what was broken, completed what was missing, and proved it with `backend/tests/pic/test_finance_journey.py` (one sequential scenario through the real HTTP API in its own tenant FIN1) and `backend/tests/pic/test_finance_controls.py` (unit tests of the TDS, duplicate-number and MSME logic).

Status legend: **VERIFIED** = covered by a named test that passes; **CODE ONLY** = implemented and read, no automated test; **MISSING** = not built. Test keys: **J01..J16** = `test_finance_journey.py`; **FC** = `test_finance_controls.py`; **R78** = `test_review_f7_f8.py`; **W1P** = `test_wave1_p2p.py`; **OO** = `test_order_ops.py`; **SIG** = `test_signature_case.py`; **R56** = `test_review_f5_f6.py`; **CON** = `test_datasets_industry_construction.py`.

1. What the specification requires

Source	Requirement
BP 16.1	Capture by portal, PDF/OCR, GST e-invoice (IRN), EDI; 2-, 3-, 4-way match, ERS; tolerances by amount, percent, price, quantity per company code; exception workflow (price, quantity, missing GR, tax, duplicate); credit/debit notes and retention; posting to ERP AP with idempotency and reconciliation; payment status, early-payment discount, SCF hooks; GRIR accruals
BP 16.2	Signature case INV-TF-8812: billed 5,400, quantity exception 20, credit note, matched, posted (ERP 5100004421); default tolerances price 1% or 500 (lower), quantity 0%, tax exact; configurable per company code and supplier group
BP 16.3	INV-01 inbox, INV-02 match and exception resolver, INV-03 posting and reconciliation, INV-04 payables insights (due dates, discounts captured, DPO, GRIR ageing)
BP 16.4	Duplicate check supplier + number + amount + date; GR before approval unless ERS; services match approved SES; posting key = tenant + company code + supplier + invoice no.; payment block until all exceptions resolved
BP 16.5	KPIs: first-pass match %, exception cycle time, duplicates prevented, discount capture %, DPO
BP 17 (M11)	Realised savings only from posted invoices; cost avoidance never counts towards budget; finance validator differs from the initiative owner
F7 (21.7)	Submitted, capture and duplicate check, 2/3/4-way match with tolerances, within tolerance?, resolve exception (credit note / approve / dispute), post to ERP (idempotent), payment run, realised savings and scorecard update, paid and reconciled; retries never double-post
TD 5/6	Invoice states Captured, Duplicate Check, Matching, Exception or Matched, Approved, Posted, Paid, Rejected; guard on Matched to Posted: no open exceptions, posting key reserved
TD 7.3 DMN-03	Price 1% or 500 per line (lower), quantity 0% vs accepted GR, quantity vs SES 0%, tax exact with IRN validated, duplicate same supplier + number or amount +-1 and date +-3 days
TD 9.4	Algorithm: PO line, accepted GR less already invoiced, price band, tax (rate, HSN/SAC, IRN valid), exceptions with suggested action, matched then approve then post (idempotent)
TD 10/11	`POST /v1/invoices`, `/match`, `/post` (Idempotency-Key), ETag/If-Match, problem+json; events invoice.captured, invoice.exception.raised, invoice.posted, payment.completed
TD 12.3	GST e-invoice IRN and signed QR validated on capture
TD 14	SoD engine SOD-01..07 at command time; bank-change fraud controls (callback, dual approval, cooling-off)
India	GST, TDS at payment, MSME 45-day payment rule (MSMED Act s.15), retention/withholding

2. Traceability: specification step to code to proof

#	Spec rule / step	Code	Proof	Status
1	F7-01 capture via portal / AP clerk / PDF (OCR)	`payables.capture`, `api/order_pay.py:capture_invoice`, `services/ocr.py`	J02, J04, J05, R78, SIG	VERIFIED
2	F7-01 GST e-invoice IRN + QR validation on capture	`services/einvoice.py:verify`, `payables.capture`	J02 (valid IRN), J05 (invalid IRN raises TAX), SIG	VERIFIED
3	F7-01 EDI INVOIC inbound	`services/edi_invoice.py`	W1P (inbound INVOIC, AERAK rollback, credit/debit syntax)	VERIFIED
4	F7-02 hard duplicate (supplier + number + FY)	unique key + `capture` 409	J02 (409 DUPLICATE_INVOICE, Idempotency-Key replay)	VERIFIED
5	F7-02 fuzzy duplicate amount +-1, date +-3 d	`payables._duplicate_check`	R78	VERIFIED
6	F7-02 OCR / formatting variants of the number	`finance_controls.normalise_invoice_no`, `_duplicate_check`	J04, FC	VERIFIED (new)
7	F7-03 3-way match vs accepted GR less invoiced, qty 0%	`payables.match`	J02 (clean), J04 (QUANTITY +50), R78, SIG	VERIFIED
8	partial receipts and several invoices on one PO line	`pending_on_line`, `over_claimed`, row lock	R78 (pendingInvoicedQty), J02/J04 (600 then 400 of 1000)	VERIFIED
9	4-way (inspection accepted), 2-way, SES, ERS match	`payables.match`	R56, R78, CON, datasets	VERIFIED (existing)
10	price tolerance min(1%, 500) per line	`payables.match`	J05 (PRICE, variance 2,000 INR against tolerance 500), R78	VERIFIED
11	tax exact (rate, HSN, tax amount, IRN) + CBIC HSN master	`payables.match`, `services/gst.py`	J05 (IRN), existing	VERIFIED
12	tolerance by company code and supplier group	`rules.resolve("DMN-03", supplier_group=...)`	W1P `test_invoice_tolerance_follows_the_supplier_group_scope`	VERIFIED
13	missing GR exception then GR posted resolution	`match`, `resolve_exception`	J06	VERIFIED
14	quantity exception resolved by credit note (value and quantity credits)	`submit_credit_note`, `resolve_exception`	J04, R78, SIG	VERIFIED
15	price exception: AP asks, buyer other than PO creator decides, PO re-priced	`_request_price_acceptance`, `_price_variance_decided`	J05	VERIFIED

#	Spec rule / step	Code	Proof	Status
16	dispute keeps the block; reject closes	`resolve_exception(DISPUTE)`, `reject`	J05, J04	VERIFIED
17	SOD-04 PO creator cannot resolve exceptions on the PO	`resolve_exception` + `sod.guard`	R78	VERIFIED
18	payment block until all exceptions resolved	`match`, `post`	J04, J05, J06, J07	VERIFIED
19	credit note netting in payment	`payment_run`	J12 (INV2 pays gross less credit note)	VERIFIED
20	debit note: match, buyer chain for charges, AP approval, ERP debit memo, nets into payment	`services/debit_notes.py`	J08, J09, W1P	VERIFIED
21	**four-eyes: capturer cannot approve (SOD-08)**	`payables.approve`, sod rule SOD-08	J04	VERIFIED (new)
22	**payer differs from approver (SOD-09)**	`payables._assess_payment`, sod rule SOD-09	J12	VERIFIED (new)
23	SOD-02/02a bank changer cannot pay / post	`_assess_payment`, `post`, `approve`	R78 (existing)	VERIFIED
24	F7-06 post to ERP, key tenant + company + supplier + number	`payables.post`, `integration.enqueue`	J03, SIG (ERP 5100004421)	VERIFIED
25	posting replay returns same ERP document	`api/order_pay.py:post_invoice`	J03	VERIFIED
26	ERP failure then retry; lost response then query-before-retry	`integration.process`, `post`	J04 (fail then admin replay), J07 (lost response, one ERP document)	VERIFIED
27	Idempotency-Key retry sees the final state (not a stale interim answer)	`Idempotent.release`, `post_invoice`	J04	VERIFIED (defect fixed)
28	ETag / If-Match on invoice writes	`order_pay._locked_invoice`	J03 (412 on stale), J14 (savings)	VERIFIED (new)
29	Idempotency-Key on credit note, exception resolve, payment run, file, bank results	`api/order_pay.py`, `api/finance.py`	J04, J09, J11	VERIFIED
30	F7-07 due date from payment terms (invoice date + net days)	`_invoice_posted`	J03 (Dec 3 + 30 = Jan 2)	VERIFIED
31	**MSME 45 days from acceptance**	`finance_controls.due_date`, `acceptance_date`	J10, J13, FC	VERIFIED (new)

#	Spec rule / step	Code	Proof	Status
32	MSME interest (3x bank rate, monthly compounding)	`finance_controls.msme_interest_minor`	FC; needs the bank rate from the tenant	VERIFIED formula, rate must be supplied
33	early-payment discount (terms 2/10 N30), SCF hooks	`services/early_payment.py`	J09, W1P	VERIFIED
34	payment run with supplier and invoice filters, idempotent	`payables.payment_run`, `api/order_pay.py:payment_run`	J09, J12, J13, R78	VERIFIED
35	**payment proposal (dry run, same logic as the run)**	`payables.payment_proposal`, `GET /v1/payments/proposal`	J09, J12, J13	VERIFIED (new)
36	blocked supplier: invoices payment-blocked, no new invoices	`supplier.block`, `capture`, `payment_run`	J07, J12	VERIFIED
37	bank-change cooling-off holds payment	`_assess_payment`	J12 (skipped with reason)	VERIFIED
38	**TDS determination (section, rate, thresholds, no-PAN, catch-up) and deduction at payment**	`finance_controls.compute_tds / at_payment`, `_fix_fx_and_tds`	J10, J13, J06, FC	VERIFIED (new)
39	TDS register (quarterly return source)	`finance_controls.tds_register`, `GET /v1/finance/tds-register`	J13	VERIFIED (new)
40	retention withheld and released	`retention_by_po`, `release_retention`	CON	VERIFIED (existing)
41	advance recovery (construction)	`services/ind_construction.py`	CON	VERIFIED (existing)
42	**foreign-currency invoice: currency from the PO, rate on the invoice date, group amount, USD payment**	`capture`, `finance_controls.fx_for`, `payment_run`	J11, J13	VERIFIED (new)
43	**posting period close and cut-off**	`finance_controls.close_period / assert_open`, `GET/POST /v1/finance/periods`	J11	VERIFIED (new)
44	payment file ISO 20022 pain.001 / NEFT, one live file per run, content audited	`services/bank_files.py`	J09, J12, OO	VERIFIED
45	**bank result per payment: credited with UTR, returned with reason, invoice payable again**	`bank_files.record_bank_results`, `payables.return_payment`	J09, J12	VERIFIED (new)

#	Spec rule / step	Code	Proof	Status
46	**returned payment's ERP clearing document listed until reversed**	`payables._reconcile_payments_and_memos` (PAYMENT_RETURN_OPEN)	J12	VERIFIED (new)
47	F7-09 payment completed only when the ERP clears	`_payment_posted`, `payment.completed`	J09, J13 (5 events)	VERIFIED
48	INV-03 nightly reconciliation (invoices, payments, credit and debit memos)	`payables.reconcile`	J12, R78, OO	VERIFIED
49	INV-04 insights: due ladder, discount, DPO, first pass, GRIR ageing, MSME, exception ageing, TDS	`payables.insights`	J13	VERIFIED
50	GRIR accrual (received not invoiced)	`payables.grir`	R78, cut-off snapshot J11	VERIFIED
51	M10 KPIs (5 of the blueprint)	`services/kpi.py` M10 section	J15	VERIFIED
52	F7-08 realised savings only from posted invoices; net of credit notes	`savings._realised`	J14 (5,000,000 = 600 + 400 pieces x 50.00)	VERIFIED
53	**realised savings net of value (price-difference) credit notes**	`savings._realised`	none (defect fixed by reading; quantity-credit path covered by J14)	CODE ONLY
54	cost avoidance never counts towards budget	`savings.waterfall`	J14	VERIFIED
55	finance validation, validator differs from owner (SOD-07)	`savings.validate`	J14 (403 non-finance, 409 SOD-07 for the owner, 412 stale, validated by finance)	VERIFIED
56	audit trail of every state change, hash chain valid	`platform/audit.py`	J15	VERIFIED
57	SoD violations leave evidence	`sod._log_violation`	J15 (SOD-07, 08, 09)	VERIFIED
58	tenant isolation (RLS)	row-level security	J16	VERIFIED
59	event schemas for new events	`events/schemas/*.json`	J15 validates payloads	VERIFIED
60	INV-01 filters (status, supplier, company, text, dates, payment, block, match, exceptions, MSME, amount)	`api/lists.py:invoices`	J16	VERIFIED (new)
61	UI: invoice list filters, payment run dialog with filters and proposal, bank results, period close, MSME/TDS/FX on the invoice	`frontend/src/app/pic/invoices/*`	`tsc --noEmit`, eslint; browser pass is done by the lead	CODE ONLY

#	Spec rule / step	Code	Proof	Status
62	**scheduled payment-due trigger: proposal for invoices due (MSME 45-day deadline included) and early-payment discount windows, approval task for the finance approver, never pays by itself**	`job_handlers.payment_due`, `_proposal_detail`, `_discount_windows`, `_proposal_decided` (job `payment-due`, rule PAYMENT-RUN)	J17, `test_wave1_jobs` (proposal, never pays with autoRun)	VERIFIED (new)
63	**supplier finance maintenance screen: MSME class, Udyam no., PAN, TDS section, s.197 certificates; ETag, audit, role gate, format validation**	`GET/PUT /v1/suppliers/{code}/finance`, `suppliers/[code]/_finance.tsx`	J18 (API: formats, 403, 412, PAN never in the audit trail); screen: tsc, eslint	VERIFIED (API), CODE ONLY (screen)
64	**ERP reversal message for a returned payment (idempotent PAYREV key, retry, replay); closes PAYMENT_RETURN_OPEN when the ERP confirms**	`payables._request_erp_reversal`, `_payment_reversed`, DOC_TYPES `PAYMENT_REVERSAL` (simulator; live connectors report it unmapped like the debit memo)	J12 (ERP down: item stays open), J20 (replay posts once, item closes)	VERIFIED (new), simulator only
65	**bank statement import: camt.053 and MT940, dry run then apply, idempotent per statement id, amount / conflict / unsent-file checks**	`services/bank_statement.py`, `POST /v1/bank-statements/import`, fixtures `tests/pic/fixtures/bank`	J20, FC (parsers, refusals)	VERIFIED (new)
66	**period close blocks credit notes, debit notes and goods receipts (problem+json with the reopen rule); GR/IR snapshot persisted per close**	`finance_controls.assert_open` in `submit_credit_note`, `debit_notes.submit/post`, `receiving.post_gr`; `grir_snapshot`	J21, J11	VERIFIED (new)
67	**s.197 lower-deduction certificate per supplier (number, rate, validity, section), applied at payment when valid**	`finance_controls.valid_certificate / at_payment_detail`, `tds_certificate`, `POST /v1/suppliers/{code}/tds-certificates`	J19 (0.5 % on invoice and note, payment keeps the certificate number), FC	VERIFIED (new)
68	**TDS on debit notes (additional taxable value, thresholds count earlier notes), netted into the payment**	`finance_controls.compute_note_tds`, `debit_notes.post`, `_assess_payment`	J19 (INR 200 at 2 % fixed at posting, INR 50 at the certificate rate), FC	VERIFIED (new)
69	**Form 26Q / 16A style CSV of TDS deducted per quarter (finance scope, audited with hash, formula-injection safe)**	`finance_controls.tds_export`, `GET /v1/finance/tds-export`	J19, FC (quarters, CSV safety)	VERIFIED (new)
70	UI: TDS export, bank statement upload, posting periods card	`invoices/_close.tsx`, `invoices/_ops.tsx`	tsc, eslint; browser pass by the lead	CODE ONLY

Counts: 70 rows; **VERIFIED 67** (22 of them new or fixed in this work; row 63 verified through the API, its screen is code-only), **CODE ONLY 3** (rows 53, 61, 70), **MISSING 0**. (Row 32 counts as verified for the formula only; row 64 is verified against the ERP simulator, live connectors report the message unmapped.)

3. What works (unchanged by this pass)

The F7 core was already enterprise grade: the 3/4-way/SES/ERS match with row-locked PO lines and over-claim re-checks, DMN-03 per company code and supplier group, value and quantity credit notes, the buyer price-variance approval, debit notes, early payment and SCF hooks, retention and advance recovery for construction, EDI INVOIC, the posting key and `integration_message` uniqueness with query-before-retry, one live payment per invoice (partial unique index), bank-change cooling-off, payment files, nightly reconciliation, the savings ledger and its SOD-07 check. Baseline before any change: `test_review_f7_f8`, `test_wave1_p2p`, `test_order_ops`, `test_signature_case`, `test_review_f5_f6`, `test_datasets_industry_construction`: 108 passed.

4. Defects found and fixed, gaps completed

#	Finding	Fix (file)
D1	**Retry with the same Idempotency-Key replayed the interim answer "POSTING" for ever** when the first post parked in RETRY (ERP down): the documented retry path never saw the posted state.	`platform/core.py` `Idempotent.release`; `api/order_pay.py` `post_invoice` releases the reservation when the result is not final
D2	AP clerk could capture an invoice and approve it herself (AP_CLERK holds capture, approve and post).	SOD-08 (`INVOICE_CAPTURE` vs `INVOICE_APPROVE`): `payables.capture` records, `approve` guards; migration 0017 inserts the rule for every tenant, `seed/reference.py` for new ones
D3	Nothing stopped the user who approved an invoice from releasing its payment.	SOD-09 (`INVOICE_APPROVE` vs `PAYMENT_APPROVE`): `payables._assess_payment` ; the invoice is skipped with the rule in the reason
D4	A debit note with only a CHARGE line could be approved by the creator of the PO it bills (SOD-04 reached price lines only).	`services/debit_notes.py` `match` excludes the creators of every order on the invoice
D5	Invoice currency was hard-coded INR at capture and Payment currency defaulted to INR: a foreign-currency PO produced an INR invoice and an INR bank line; no exchange rate anywhere on the invoice.	`payables.capture` takes the PO currency (422 for mixed currencies), `_fix_fx_and_tds` fixes the rate at first post (409 FX_RATE_MISSING), `payment_run` pays in the invoice currency; insights convert to the company currency
D6	Realised savings treated a value (price-difference) credit note as a quantity credit, understating the saving; the initiative lookup raised `MultipleResultsFound` when two initiatives shared a material.	`services/savings.py` `_realised` uses the match's own credit helpers and takes the oldest initiative
D7	Duplicate detection ignored OCR and formatting variants (INV-0012, inv 12, INVO12).	`finance_controls.normalise_invoice_no` + `payables._duplicate_check` hold the variant (DUPLICATE, `variant: NUMBER_NORMALISED`)
D8	MSME 45-day rule absent.	migration 0017 (supplier.msme_class, udyam_no; invoice.acceptance_date, msme_capped), rule set MSME, `finance_controls.due_date` , payment proposal ordering, insights
D9	TDS was a column filled only by the SAP loader; nothing computed or reported it.	rule set TDS, supplier.tds_section, `finance_controls.compute_tds` at posting (fixed once so retries send the same payload), `at_payment` , `payment.tds_minor` , ERP payload, register, ETag-guarded supplier maintenance route
D10	Bank acknowledgement was file-level only: a payment the bank returned left the invoice PAID for ever; no UTR.	`bank_files.record_bank_results` , `payables.return_payment` , state `PAID` to `POSTED` , payment `bank_status` , reconciliation item for the ERP reversal
D11	No posting-period control or cut-off.	table `posting_period` (RLS), `finance_controls.close_period/reopen_period/assert_open` , checks in `post` and the payment run, GR/IR snapshot, events finance.period.closed / reopened
D12	No payment proposal; the run could only be executed.	`payables.payment_proposal` shares `_assess_payment` with the run

#	Finding	Fix (file)
D13	Invoice writes had no ETag / If-Match; credit-note and resolve had no Idempotency-Key; worklist filtered by status only; no payment list.	`api/order_pay.py` (`_locked_invoice`, ETag on GET), `api/lists.py` (12 filters, one grouped query for open exceptions), `api/finance.py` (`GET /v1/payments`)
D14	Skipped list of the run keyed by invoice number only (a debit-note payment and its invoice collide in the UI).	frontend key includes the position

Second pass (closing the gaps of section 5):

#	Finding	Fix (file)
D15	The `payment-due` job existed but could pay by itself with `autoRun`, and its proposal showed gross totals only (no MSME deadline, TDS, discounts, no discount windows).	`services/job_handlers.py`: autoRun is ignored by policy (reported as `autoRunIgnored`), the task evidence now carries the full proposal (amounts net of credit notes, TDS and discount, MSME flags, skipped reasons) and the invoices whose terms discount closes within the horizon; approving takes the terms discount and runs the normal payment run. `test_wave1_jobs` was changed accordingly (the old test asserted the auto-run).
D16	A returned payment left its ERP clearing document to be reversed by hand.	`PAYMENT_REVERSAL` message (key `PAYREV
D17	Bank results were keyed by hand.	`services/bank_statement.py` (camt.053 with a DTD/entity refusal, MT940), analysis states READY / UNCHANGED / CONFLICT / AMOUNT_MISMATCH / FILE_NOT_SENT / UNMATCHED / AMBIGUOUS / NO_REFERENCE, apply through `bank_files.record_bank_results`, table `bank_statement_import` (unique per tenant and statement id).
D18	Credit notes, debit notes and goods receipts could be posted into a closed period; no GR/IR evidence at close.	`finance_controls.assert_open` in the three paths; `PERIOD_CLOSED` problem carries `period`, `companyCode`, `reopenScope` and the reopen rule; table `grir_snapshot` (one row per close, line by line).
D19	No lower-deduction certificate, no TDS on debit notes, no return file.	tables `tds_certificate`; columns `debit_note.tds_*`, `payment.tds_base_minor / tds_rate / tds_certificate`; `finance_controls`; export endpoint.
D20	PAN could not be maintained; Udyam had no format check.	`PUT /v1/suppliers/{code}/finance` accepts `pan` (must be the one inside the GSTIN, stored as hash, only the masked value is audited) and validates `UDYAM-XX-00-0000000`.

Migration 0017 was extended (it is local only): `debit_note.tds_base_minor / tds_minor`, `payment.tds_base_minor / tds_rate / tds_certificate / reversal_doc`, tables `grir_snapshot`, `tds_certificate`, `bank_statement_import` (grants and forced RLS). New files: `services/bank_statement.py`, `tests/pic/fixtures/bank/*`, `frontend/.../invoices/_close.tsx`, `frontend/.../suppliers/[code]/_finance.tsx`.

Shared files changed backward compatibly: `payables.py`, `bank_files.py`, `savings.py`, `debit_notes.py`, `platform/core.py` (new method), `platform/auth.py` (+ scope `period:close` for FINANCE_CONTROLLER), `platform/roles.py`, `seed/reference.py` (+ SOD-08/09, rule sets MSME and TDS), `events/catalogue.py`. New: migration `0017_finance_complete.py` (down_revision 0016), `models/finance_close.py`, `services/finance_controls.py`, `platform/finance_defaults.py`, `api/finance.py`, four event schemas.

5. What remains

- Live ERP reversal: `PAYMENT_REVERSAL` is mapped for the simulator only (like the debit memo and SCF types); SAP S/4, Dynamics, Tally and Odoo connectors report it unmapped, so with them the treasury clerk still reverses the clearing document and resolves the reconciliation item.
- The payment-due job proposes in the tenant time zone once a day; it does not chase a proposal nobody decides (the task SLA escalation of the workflow engine does).
- TDS: certificates are per supplier, rate and validity (no amount cap of the certificate); no Form 16A PDF, no challan/BSR matching, no TCS, no TDS on advances, no foreign-remittance TDS (s.195). The PAN of the return comes from the GSTIN (a supplier without GSTIN is exported as PANNOTAVBL). The default sections remain illustrative.
- MSME: interest is computed only when the tenant publishes the bank rate; no accrual posting and no 15-day rule for suppliers without a written agreement.
- FX: rate fixed at invoice date, no revaluation or realised gain/loss at payment; foreign payment files carry no SWIFT data.
- Bank statements: camt.053 and a basic MT940 (one :61:/:86: pair per entry, matching by payment number); no CAMT.054, no multi-payment batch entries (reported AMBIGUOUS), no automatic matching of other bank movements.
- SOD-08 is recorded for supplier invoice capture; consignment settlements and retention releases (system-built AP documents) do not record a capturer.
- Realised savings from value credit notes (row 53) has no automated test.
- The browser pass of the new screens (invoice list, payment run, bank results and statement upload, posting periods, TDS export, supplier finance card) is done by the lead.

6. What the product owner must supply

Item	Why
Tax team sign-off of TDS sections, rates, thresholds and the section codes of the Act in force from 1 April 2026; publish as rule set TDS (`POST /v1/admin/rules`)	the shipped defaults are illustrative
PAN and TDS section per supplier (migration or screen), lower-deduction certificates	determines what is withheld
RBI bank rate (rule MSME `bankRatePct`) and the position on suppliers without a written payment agreement	MSME interest
Udyam verification source and the list of micro/small suppliers	MSME class
Debtor bank account per company code (rule PAYMENT-FILE: debtorName, debtorAccountRef, debtorIfsc), bank file specification, bank acknowledgement / statement format and host-to-host access (`PIC_BANK_SFTP_*`)	payment files, automatic bank results
ERP posting profile for TDS, MSME interest, credit memos and payment reversal; SAP (or other) sandbox access	live posting and reversal
Exchange-rate source (daily reference rates) loaded into `fx_rate`	foreign-currency invoices
Fiscal calendar, who may close and reopen periods, and the cut-off day	period control
IRP / GSP credentials (`PIC_IRP_*`)	live e-invoice verification
Invoice approval value bands and the finance approvers	approval policy
Sample camt.053 and MT940 files from the bank (the shipped fixtures are synthetic) and whether the bank echoes our payment number as EndToEndId / reference	statement matching
s.197 certificates of the suppliers (number, rate, validity) and the tax team's rule on certificate amount caps	lower deduction
Live ERP message for reversing a payment (SAP reversal BAPI / OData service, Dynamics, Tally, Odoo)	closing returned payments automatically on live systems
Time of day and approver role for the payment proposal (rule PAYMENT-RUN: horizonDays, approverRole, slaBusinessDays)	scheduled trigger

7. How to run the proof

```
backend\venv\Scripts\python.exe <scratchpad>\mkdb.py pic_test_fin
set TEST_DATABASE_URL=postgresql://postgres@127.0.0.1:54329/pic_test_fin
backend\venv\Scripts\python.exe -m pytest tests/pic/test_finance_journey.py tests/pic/test_finance_controls.py -q
```

Developers iterating on a seeded database add `PIC_TEST_KEEP_DB=1` and a fresh tenant code (`FIN_TEST_TENANT=FIN2`). New environment variables: none (existing `PIC_REQUIRE_IF_MATCH` now also applies to invoice and supplier-finance writes).